

NMAM INSTITUTE OF TECHNOLOGY, NITTE
(An Autonomous Institution affiliated to VTU, Belagavi)
Sixth Semester B.E. (AIML) (Credit System) Degree Examinations
May - June 2023

20AM601 - NEURAL NETWORKS AND DEEP LEARNING

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.*

Unit – I		Marks	BT*	CO*	PO*
1.	a) Derive the chain rule for ANN, using minimum 2 hidden layers.	8	L*3	1	2
	b) Explain dead ReLU problem. How can it be solved using Leaky ReLU activation function?	6	L2	1	1
	c) Describe the forward propagation process in ANN with a neat diagram.	6	L2	1	1
2.	a) Illustrate the working of a perceptron.	8	L2	2	1
	b) Discuss vanishing gradient problem with an example using weight update formula.	7	L2	2	1
	c) Compare L1 & L2 regularization techniques.	5	L2	2	1
3.	a) Demonstrate AdaGrad optimizer and derive the weight updation formula.	6	L3	2	1
	b) Illustrate with an example, back propagation algorithm.	8	L3	1	1
	c) Discuss about different weight initialization techniques used in Artificial Neural Networks.	6	L2	2	1
Unit – II					
4.	a) Justify the need of RNN.	4	L2	4	1
	b) Demonstrate the padding process by considering 5*5 image input for the filter size of 3*3 horizontal and vertical edge detection. Explain the feature map formula with padding.	8	L3	3	2
	c) Discuss the problems with encoder and decoder circuit.	8	L3	4	1
5.	a) Elaborate LSTM recurrent neural network.	6	L2	4	2
	b) Discuss the need of data augmentation and explain the methods of it.	8	L3	4	1
	c) Compare Artificial Neural Network and Convolutional Neural Network.	6	L2	3	1
6.	a) Summarize the different architectures of Convolutional Neural Network.	8	L3	3	1
	b) Write a program which uses defragmentation process in CNN.	7	L3	4	1
	c) Discuss the disadvantages of strides with an example.	5	L2	3	1
Unit – III					
7.	a) Discuss the process of application of deep learning in Computer Vision.	10	L2	5	1
	b) Elaborate how to identify, whether to gather more data in deep learning.	10	L2	5	1
8.	a) Illustrate the applications of deep learning in Natural language processing.	10	L2	5	1
	b) Discuss different baseline models present in neural network and deep learning.	10	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
